The stated Stern-square normalization tends to zero

Disproof of Conjecture 00000000228

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Abstract

For the actual Stern diatomic sequence, the entire sequence $s(n^2)/3^{\log_2 n}$ tends to zero as $n \to \infty$. An exact three-bit recurrence bound gives the geometric estimate $25(25/27)^m$ on every block $8^m \le n < 8^{m+1}$. The genuine uniform-index empirical distributions therefore converge to the point mass at zero, not to a continuous fractal law. The proof handles all positive indices, the original real logarithmic power and actual probability measures.

The original sequence and its adjacent pairs

Use exactly the defining rules from the source:

$$s(0) = 0, \quad s(1) = 1, \quad s(2n) = s(n), \quad s(2n+1) = s(n) + s(n+1).$$

Let $v_n = (s(n), s(n+1))$. The rules give

$$v_{2n} = L(v_n), \qquad v_{2n+1} = R(v_n), \qquad L(a,b) = (a, a+b), \quad R(a,b) = (a+b, b).$$

These formulas, starting at $v_0 = (0,1)$, construct the actual sequence by repeated division of the index by two. The Lean project proves all four original rules, the adjacent-second-coordinate identity, and uniqueness among sequences satisfying those rules. It also proves $s(n) > 0$ for every $n > 0$.

An exact three-bit growth bound

If $0 \le a, b \le B$, every composition of three maps from $\{L, R\}$ has both coordinates at most $5B$. Explicitly, writing $v_k = (a, b)$, one obtains

r	v_{8k+r}
0	$(a, 3a+b)$
1	$(3a+b, 2a+b)$
2	$(2a+b, 3a+2b)$
3	$(3a+2b, a+b)$
4	$(a+b, 2a+3b)$
5	$(2a+3b, a+2b)$
6	$(a+2b, a+3b)$
7	$(a+3b, b)$

Every displayed coordinate has nonnegative coefficients with sum at most five. The bound is algebraic and valid for arbitrary k, not inferred from finite testing.

A global bound at every index

Induction on j, dividing an index by eight at each step, now proves

$$0 \le n \le 8^j \quad \Longrightarrow \quad s(n) \le 5^j, \qquad s(n+1) \le 5^j.$$

At $j = 0$ the indices are zero and one, with pairs $(0, 1)$ and $(1, 1)$. For the induction step, $n \le 8^{j+1}$ gives $\lfloor n/8 \rfloor \le 8^j$; three binary steps multiply the pair bound by at most five. The endpoint is included throughout.

In particular, if $n \le 8^{m+1}$, then $n^2 \le 8^{2(m+1)}$, so

$$s(n^2) \le 5^{2(m+1)} = 25^{m+1}.$$

On the other hand, if $8^m \le n$, monotonicity of the ordinary logarithm gives

$$\log_2 n \ge 3m, \qquad 3^{\log_2 n} \ge 27^m.$$

Consequently for every positive index in the block,

$$8^m \le n \le 8^{m+1} \quad \Longrightarrow \quad 0 < \frac{s(n^2)}{3^{\log_2 n}} \le \frac{25^{m+1}}{27^m} = 25 \left(\frac{25}{27} \right)^m.$$

For each $n \ge 1$, take $m = \lfloor \log_8 n \rfloor$. These blocks cover every positive index, and $m \to \infty$ with n. Since $25/27 < 1$, we obtain the full-sequence limit

$$a_n := \frac{s(n^2)}{3^{\log_2 n}} \longrightarrow 0.$$

The denominator here is exactly $3^{\log n / \log 2}$, not a rounded exponent or a replacement normalization. At $n = 1$ it is one; the undefined source index $n = 0$ is never sampled.

The actual empirical distribution

For sample count $N \ge 1$, let

$$\nu_N = \frac{1}{N} \sum_{n=1}^{N} \delta_{a_n}.$$

This is the genuine pushforward of a uniform random index in $\{1, \ldots, N\}$ through the actual normalized sequence. All a_n are positive. Thus its CDF $F_N(x) = \nu_N((-\infty, x])$ is exactly zero for every $x \le 0$ and every N.

Degeneration and exclusion of a continuous limit

Fix $x > 0$. Since $a_n \to 0$, there is K such that $a_n \le x$ for every $n \ge K$. At most K of the first N observations can exceed x. Therefore

$$0 \le \nu_N((x,\infty)) \le \frac{K}{N} \longrightarrow 0, \qquad F_N(x) \longrightarrow 1.$$

We have proved the actual CDF limits

$$F_N(x) \longrightarrow \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

These are precisely the CDF values of the actual Dirac measure δ_0 at every continuity point of that CDF. In the usual continuity-point criterion, the empirical laws converge weakly to δ_0. No convergence of CDFs to the Dirac value at its discontinuity zero is asserted; indeed $F_N(0) = 0$ for every finite N.

There cannot instead be a continuous limiting CDF F. Convergence in law to such a target requires $F_N(x) \to F(x)$ at every real x. It would give $F(0) = 0$ and $F(x) = 1$ for every $x > 0$. Continuity along $x_j = 1/(j+1) \downarrow 0$ would then force $F(0) = 1$, a contradiction.

The necessary continuity-point condition can also be seen directly from the bounded-continuous-function definition of weak convergence: approximate the indicator of $(-\infty, x]$ from above and below by continuous linear ramps within distance ε of x, pass to the limit in their integrals, then let $\varepsilon \downarrow 0$ at a continuity point of the proposed CDF.

Theorem 1. *For Stern's actual diatomic sequence, $a_n = s(n^2)/3^{\log_2 n} \to 0$. Its uniform positive-index empirical laws have the point mass at zero as their weak limit and cannot have a continuous limiting CDF.*

Proof. The exact three-bit block bound gives the full-sequence limit. The finite exceptional-index estimate gives the actual CDF limits, and continuity yields the displayed contradiction for every continuous target. □

This refutes the source's proposed continuous fractal distribution under its supplied normalization. The separate informal spectral-dimension/turning-density clause need not be assigned an interpretation. No claim is made about other normalizations of Stern values.

Formalization and reproducibility

The Lean sequence is constructed by an actual terminating adjacent-pair recursion. All original recurrence rules and their uniqueness are proved. All eight three-step growth cases, the inclusive power bound, square-index bound and original real logarithmic denominator comparison are formal theorems. Mathlib's natural base-eight logarithm supplies blocks covering every positive integer, and its index tends to infinity. A geometric squeeze proves the full normalized limit.

The empirical PMF samples an actual uniform $\mathrm{Fin}(N+1)$ index and maps it to a_{i+1}. Thus the formal parameter $N \geq 0$ represents sample count $N+1$, always using positive indices. Its canonical probability measure is used for all subsequent CDF and tail expressions.

The tail probability equals the actual number of bad indices divided by the sample count. An explicit finite-set injection bounds that number by K. The exact nonpositive CDF, positive-threshold limits, equality with Mathlib's actual CDF, and continuity-point limits to the actual Dirac measure are proved. A separate theorem rules out every continuous pointwise CDF limit. The project proves these CDF statements directly; it does not claim to formalize the full general equivalence with Mathlib's weak convergence framework. The necessary condition used for the disproof is explained on the preceding page.

The supplementary `verify.py` uses independent direct-recursion and binary-pair algorithms, checks all eight block identities at 10,001 base indices, and checks the square bound at the first 10,000 positive indices, with exact integer/rational arithmetic. These finite checks are not a Lean computation oracle; the formal proof supplies the all-index result.

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Fresh-build logs, source and hashes are included. Printed dependencies contain only standard foundational axioms. Compile with `tectonic main.tex`. The author agent and parent reviewed the work internally; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000000228. Exact bilingual source included. All sequence-specific bounds are proved above and in the accompanying Lean project.